Hospital Weekend Effect Analysis

ETL Pipeline and Business Intelligence Dashboard Using MIMIC-III Clinical Database Demo

KEY INSIGHTS

- Weekend admissions account for 25% of all admissions.
- Mortality rate is 4 percentage points lower on weekends.
- Weekend patients stay longer in hospital (10.6 vs 9.4 days).
- ICU LOS is lower during weekends (4.1 vs 5.2 days).

Weekend Effect Analysis in Hospital Admissions

Comparative Analysis of Weekend and Weekday Admissions Based on Mortality, Length of Stay and ICU Utilization

Gender

F | M

Total Admissions
120

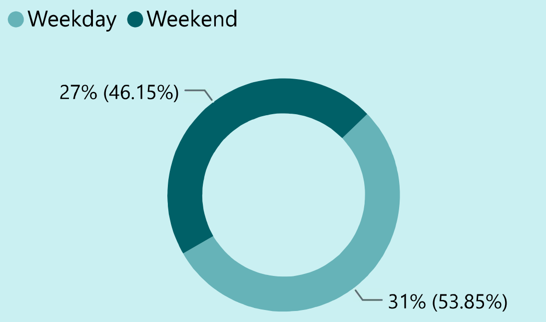
Weekend Admissions
30

Weekday Admissions
90

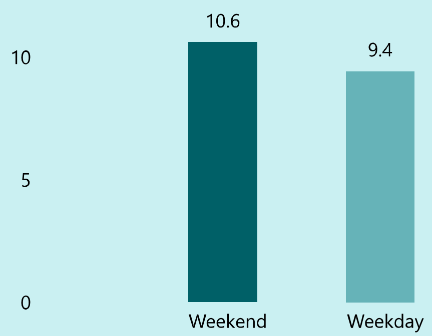
Overall Mortality Rate
30%

Avg Hospital LOS
9.72

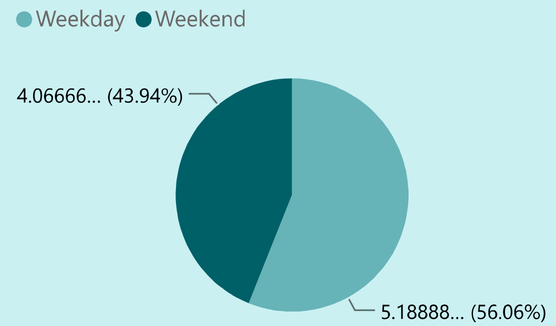
Mortality Rate Comparison



Average Length of Stay



ICU Length of Stay



Mortality Difference

-4%

Weekend mortality is 4 percentage points lower than weekday mortality in the analyzed admissions dataset.

Average LOS Difference

1.18

Weekend patients stayed 1.2 days longer on average than weekday patients.

ICU LOS Difference

-1.12

Weekend patients spent 1.1 fewer days in ICU on average than weekday patients.

Patient Population & Admission Characteristics

Understanding the demographic composition and admission pathways of hospitalized patients.

Gender

F | M

Average Patient Age
68.94

Male Population
66

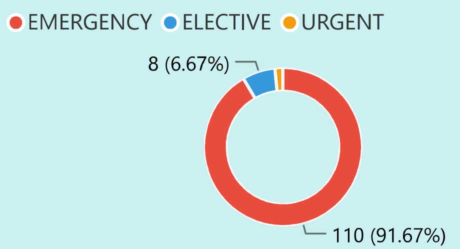
Female Population
54

Weekend Admission Share
25%

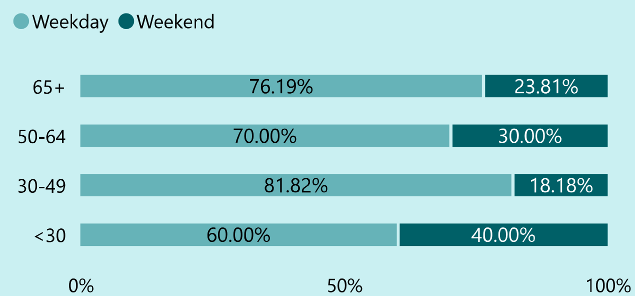
Emergency Admission Share %
91.7%

Admission Type

Emergency admissions dominate hospital intake.

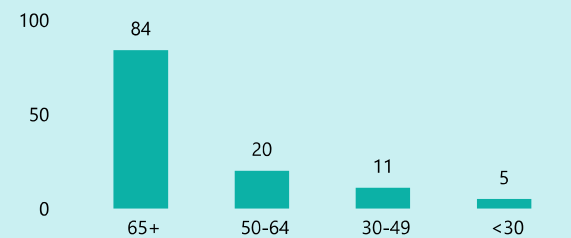


Weekend vs Weekday Age Mix



Age Profile of Admitted Patients

Admissions concentrated among older age groups.



Top Admission Locations

admission_location	Total Admissions
CLINIC REFERRAL/PREMATURE	12
EMERGENCY ROOM ADMIT	73
TRANSFER FROM HOSP/EXTRAM	24

Top Discharge Type

discharge_location	Total Admissions
SNF	37
DEAD/EXPIRED	36
HOME	14